

Problem

Repeat Example 12.13 for the network in Fig. 12.24 (see Practice Prob. 12.9). *Hint* Connect the reference point o in Fig. 12.33 to point B .

Example 12.13:

Three wattmeters W_1 , W_2 , and W_3 are connected, respectively, to phases a , b , and c to measure the total power absorbed by the unbalanced wye-connected load in Example 12.9 (see Fig. 12.23). (a) Predict the wattmeter readings. (b) Find the total power absorbed.

Solution:

Part of the problem is already solved in Example 12.9. Assume that the wattmeters are properly connected as in Fig. 12.36.

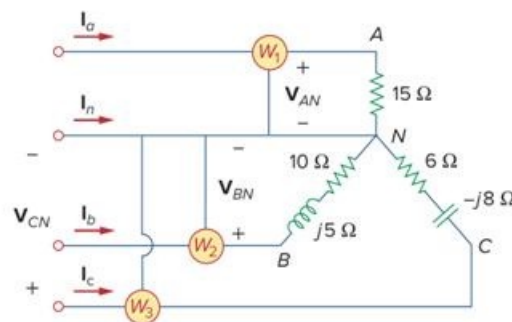


Figure 12.36
For Example 12.13.

(a) From Example 12.9,

$$\mathbf{V}_{AN} = 100 \angle 0^\circ, \quad \mathbf{V}_{BN} = 100 \angle 120^\circ, \quad \mathbf{V}_{CN} = 100 \angle -120^\circ \text{ V}$$

while

$$\mathbf{I}_a = 6.67 \angle 0^\circ, \quad \mathbf{I}_b = 8.94 \angle 93.44^\circ, \quad \mathbf{I}_c = 10 \angle -66.87^\circ \text{ A}$$

We calculate the wattmeter readings as follows:

$$\begin{aligned} P_1 &= \operatorname{Re}(\mathbf{V}_{AN} \mathbf{I}_a^*) = V_{AN} I_a \cos(\theta_{V_{AN}} - \theta_{I_a}) \\ &= 100 \times 6.67 \times \cos(0^\circ - 0^\circ) = 667 \text{ W} \end{aligned}$$

$$\begin{aligned} P_2 &= \operatorname{Re}(\mathbf{V}_{BN} \mathbf{I}_b^*) = V_{BN} I_b \cos(\theta_{V_{BN}} - \theta_{I_b}) \\ &= 100 \times 8.94 \times \cos(120^\circ - 93.44^\circ) = 800 \text{ W} \end{aligned}$$

$$\begin{aligned} P_3 &= \operatorname{Re}(\mathbf{V}_{CN} \mathbf{I}_c^*) = V_{CN} I_c \cos(\theta_{V_{CN}} - \theta_{I_c}) \\ &= 100 \times 10 \times \cos(-120^\circ + 66.87^\circ) = 600 \text{ W} \end{aligned}$$

(b) The total power absorbed is

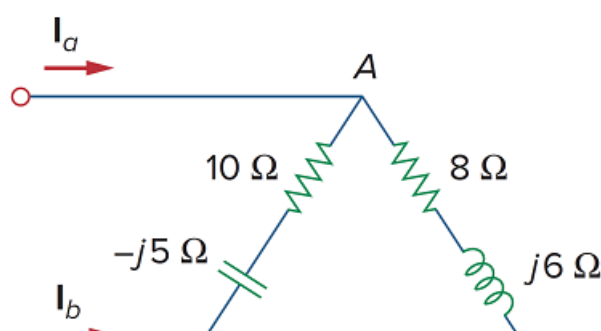
$$P_T = P_1 + P_2 + P_3 = 667 + 800 + 600 = 2067 \text{ W}$$

We can find the power absorbed by the resistors in Fig. 12.36 and use that to check or confirm this result

$$\begin{aligned} P_T &= |I_a|^2(15) + |I_b|^2(10) + |I_c|^2(6) \\ &= 6.67^2(15) + 8.94^2(10) + 10^2(6) \\ &= 667 + 800 + 600 = 2067 \text{ W} \end{aligned}$$

which is exactly the same thing.

Figure 12.23



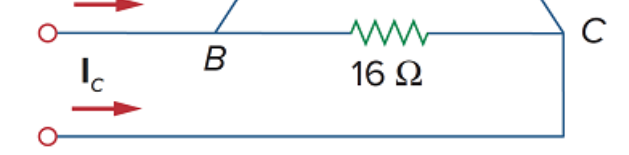
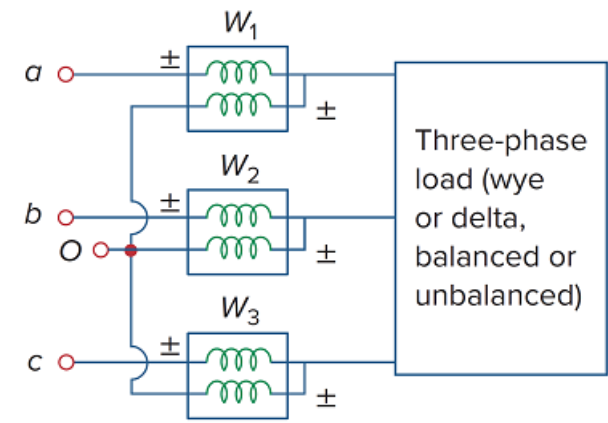


Figure 12.33



Step-by-step solution

Step 1 of 6

Draw the circuit diagram:

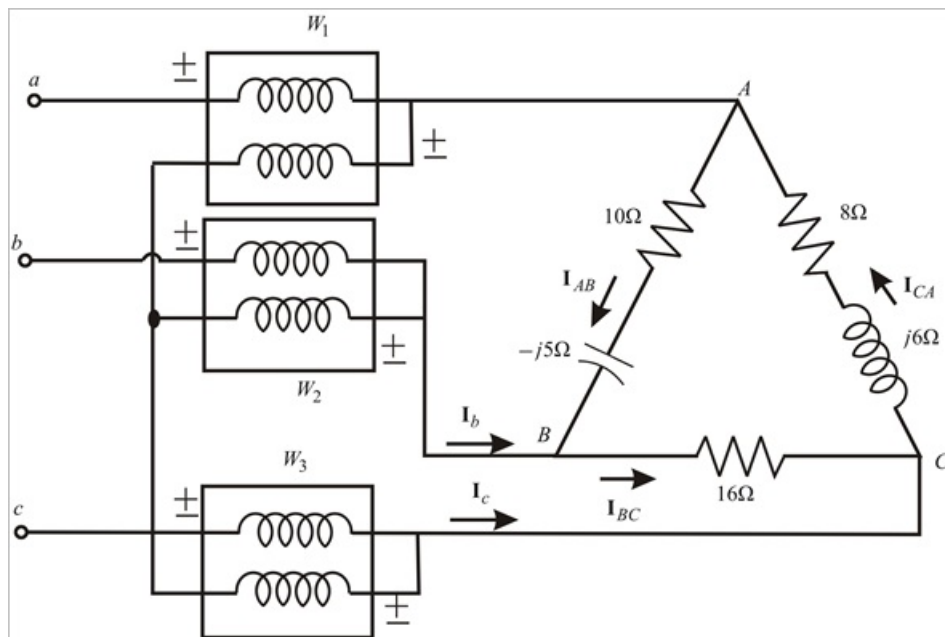


Figure 1

Step 2 of 6

Consider the following data:

Line-to-line voltage is 440 V and positive sequence

Calculate the phase voltage in delta connected load.

$$\mathbf{V}_{AB} = 440\angle 0^\circ \text{ V}$$

$$\begin{aligned}\mathbf{V}_{BC} &= \mathbf{V}_{AB}\angle -120^\circ \\ &= 440\angle (0^\circ - 120^\circ) \text{ V} \\ &= 440\angle -120^\circ \text{ V}\end{aligned}$$

$$\begin{aligned}\mathbf{V}_{CA} &= \mathbf{V}_{AB}\angle 120^\circ \\ &= 440\angle (0^\circ + 120^\circ) \text{ V} \\ &= 440\angle 120^\circ \text{ V}\end{aligned}$$

Step 3 of 6

Calculate the phase currents.

$$\begin{aligned}\mathbf{I}_{AB} &= \frac{\mathbf{V}_{AB}}{\mathbf{Z}_{AB}} \\ &= \frac{440\angle 0^\circ}{10 - j5} \\ &= \frac{440\angle 0^\circ}{11.18\angle -26.56^\circ} \\ &= 39.35\angle 26.56^\circ \text{ A}\end{aligned}$$

$$\begin{aligned}\mathbf{I}_{BC} &= \frac{\mathbf{V}_{BC}}{\mathbf{Z}_{BC}} \\ &= \frac{440\angle -120^\circ}{16} \\ &= 27.5\angle -120^\circ \text{ A}\end{aligned}$$

$$\begin{aligned}\mathbf{I}_{CA} &= \frac{\mathbf{V}_{CA}}{\mathbf{Z}_{CA}} \\ &= \frac{440\angle 120^\circ}{8 + j6} \\ &= \frac{440\angle 120^\circ}{10\angle 36.86^\circ} \\ &= 44\angle 83.12^\circ \text{ A}\end{aligned}$$

Step 4 of 6

Apply Kirchhoff's current law at node 'A', to obtain the line current $\mathbf{I}_a$.

$$\mathbf{I}_a + \mathbf{I}_{CA} = \mathbf{I}_{AB}$$

$$\begin{aligned}\mathbf{I}_a &= \mathbf{I}_{AB} - \mathbf{I}_{CA} \\ &= 39.35 \angle 26.56^\circ - 44 \angle 83.12^\circ \\ &= 35.19 + j17.59 - 5.27 - j43.16 \\ &= 29.926 - j26.08\end{aligned}$$

$$= 39.7 \angle 41.08^\circ \text{ A}$$

Apply Kirchhoff's current law at node 'B', to obtain the line current $\mathbf{I}_b$.

$$\mathbf{I}_b + \mathbf{I}_{AB} = \mathbf{I}_{BC}$$

$$\begin{aligned}\mathbf{I}_b &= \mathbf{I}_{BC} - \mathbf{I}_{AB} \\ &= 27.5 \angle -120^\circ - 39.35 \angle 26.56^\circ \\ &= -13.75 - j23.81 - 35.19 - j17.59 \\ &= -48.94 - j41.74\end{aligned}$$

$$= 64.11 \angle -139.76^\circ$$

$$= 64.11 \angle -139.76^\circ \text{ A}$$

Apply Kirchhoff's current law at node 'C', to obtain the line current $\mathbf{I}_c$.

$$\mathbf{I}_c + \mathbf{I}_{BC} = \mathbf{I}_{CA}$$

$$\begin{aligned}\mathbf{I}_c &= \mathbf{I}_{CA} - \mathbf{I}_{BC} \\ &= 44 \angle 83.12^\circ - 27.5 \angle -120^\circ \\ &= 5.27 + j43.16 + 13.75 + j23.81 \\ &= 19.02 + j67.49\end{aligned}$$

$$= 70.12 \angle 74.26^\circ \text{ A}$$

Step 5 of 6

(a)

Calculate the Watt meters reading.

$$\begin{aligned}P_1 &= \text{Re}(\mathbf{V}_{AB} \mathbf{I}_a^*) \\ &= V_{AB} I_a \cos(\theta_{AB} - \theta_{I_a}) \\ &= (440)(39.7) \cos(0^\circ - 41.08^\circ) \\ &= 13.17 \text{ kW}\end{aligned}$$

$$\begin{aligned}P_2 &= \text{Re}(\mathbf{V}_{bB} \mathbf{I}_b^*) \\ &= V_{bB} I_b \cos(\theta_{bB} - \theta_{I_b}) \\ &= 0 \times 64.11 \cos(0^\circ + 139.76^\circ) \\ &= 0 \text{ W}\end{aligned}$$

$$\begin{aligned}P_3 &= \text{Re}(\mathbf{V}_{BC} \mathbf{I}_c^*) \\ &= V_{BC} I_c \cos(\theta_{BC} - \theta_{I_c}) \\ &= 440 \times 70.12 \cos(-120^\circ + 105.73^\circ) \\ &= 29.91 \text{ kW}\end{aligned}$$

Therefore, the wattmeter readings are

$$\boxed{\begin{aligned}P_1 &= 13.17 \text{ kW} \\ P_2 &= 0 \text{ W} \\ P_3 &= 29.91 \text{ KW}\end{aligned}}.$$

Step 6 of 6

(b)

Calculate the total power absorbed.

$$\begin{aligned}P_T &= P_1 + P_2 + P_3 \\&= 13.17 + 0 + 29.91 \\&= 43.08 \text{ kW}\end{aligned}$$

Therefore, the total power absorbed is $P_T = 43.08 \text{ kW}$.